

open up new markets for the LED lighting industry.

According to 57 per cent of the survey participants, retrofit installations are the most in demand among all installation types (Fig. 2).

GST: Boon or bane?

According to survey participants, the possible impact of GST on the LED lighting industry is not yet clear. However, their opinions are summarised below.

Pros

1. A unified tax structure may help to create a unified market across the country
2. A unified tax structure may encourage big industries to support the formation of manufacturing clusters to avoid logistics costs, which were earlier nullified by differences in taxes across the country
3. Non-tax payers, which account for 30-35 per cent of the LED industry, have been enjoying an unethical price advantage compared to the big brands. GST implementation will compel non tax-compliant players to adhere to the tax system. This will have a long-term positive impact on the market and also give genuine brands a fair advantage
4. Less paperwork and borderless transits may bring down costs and delivery timelines, creating benefits across the value chain.
5. GST will improve the efficiency of the supply chain across industries, including the LED lighting sector

Cons

1. The GST rate on LED lights or fixtures, including LED lamps, is 12 per cent, whereas components and raw materials used for manufacturing LED lamps attract GST rates ranging from 18 to 28 per cent. This anomaly may result in huge revenue leakage, and might even lead to unethical accounting practices like under-invoicing
2. Metal-core printed circuit boards (MCPCB) and LED fixtures

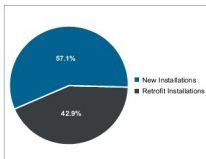


Fig. 2: Demand for LED lighting installation type; figures denote the percentage of survey participants who feel a particular LED installation type will be in great demand

will be roughly charged 12 per cent GST. This may discourage local assembly, resulting in increased fixture imports

3. Some of the imported raw materials, which have been sourced from countries under category C of the Foreign Trade Policy 2015-20, were earlier attracting tax at two per cent, but after GST implementation, these will come under the 28 per cent tax slab. This will make local manufacture of products that use these materials unviable. Similarly, most electronic components fall under the 18 per cent category, up from six per cent earlier.

4. Most SMEs avail services of unregistered dealers. Earlier, such services were not taxable as the turnover of unregistered dealers tended to be very low. After GST implementation, companies are bound to pay 18 per cent tax (as reverse GST) even in the case of services offered by non-GST registered vendors

Emerging technology trends

LED lighting not only lowers energy consumption levels but also enhances the overall lighting experience with respect to control, monitoring/sensing and connectivity, coupled with the convenience of long life and improved aesthetics. Survey participants shared insights on technology trends that will shape the LED lighting mar-

ket in India. Here is a collation of their views.

Smart connected LED lights to be the next big thing. Lighting systems will get smarter, as the possibility of autonomous, self-commissioning illumination systems is emerging. The industry has transformed from analogue to digital as LED lighting allows users to control, monitor and measure lighting output. This transformation is taking place across public, home and professional lighting, and smart connected LED lights will emerge as the largest Internet of Things (IoT) device segment within the next five to ten years. Control devices, dimmers and wireless lighting with advanced sensors will cater to the needs of modern consumers.

Driverless low-voltage direct-current (LVDC) operated products. LVDC products enable energy saving by reducing AC-DC current conversion losses. This also makes the products compatible with solar photovoltaic systems, helping them to run as LED-solar hybrid systems, which will be quite effective in India, especially in rural areas.

Advanced manufacturing practices. Shift in manufacturing techniques from through-hole to surface-mount technology (SMT) will enhance efficiency and automation while reducing operational costs. This, in turn, will help achieve break-even point quickly, in spite of a relatively higher capex cost.

Evolved on-board technology. On-board technology will evolve through the use of IC-based drivers to support touch-based technology, which is the need of the hour for smart lighting systems. This will also reduce the number of components used in drivers. Use of fewer components will reduce costs while enhancing the efficiency of the final product.

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